THE STATE UNIVERSITY OF ZANZIBAR (SUZA)

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

Software Requirements Specification (SRS)

Document Information

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1 Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the requirements for the University Innovation Connect (UIC) system.

The purpose of this system is to provide a centralized platform where students can submit innovative ideas, receive feedback from lecturers, collaborate with others, and improve their ideas into practical solutions that address real community challenges.

This document is intended for project supervisors, developers, testers, lecturers, and students involved in the development and use of the system.

1.2 Project Scope

University Innovation Connect (UIC) is a web-based platform developed for The State University of Zanzibar (SUZA).

The system allows students to submit innovative ideas, discuss them with peers, receive lecturer guidance, and improve them into meaningful projects. The platform also helps lecturers identify promising ideas and provide mentorship to students.

The system aims to strengthen innovation, collaboration, and problem-solving within the university community.

1.3 Definitions and Acronyms

Term	Definition
SRS	Software Requirements Specification
UI	User Interface
API	Application Programming Interface
UIC	University Innovation Connect
DBMS	Database Management System
Admin	System Administrator
CRUD	Create, Read, Update and Delete

1.4 References

1. Project Proposal Guidelines
2. React.js Documentation
3. Agile Software Development Methodology

2 Overall Description

2.1 Product Perspective

University Innovation Connect (UIC) is a standalone web application that supports innovation activities within SUZA.

The system acts as a bridge between students who have innovative ideas and lecturers who can provide mentorship and guidance.

The platform helps transform ideas into practical solutions through feedback, discussion, and continuous improvement.

2.2 Product Features

- User Registration
- User Login
- Idea Submission
- Idea Viewing
- Idea Editing
- Commenting on Ideas
- Voting for Ideas
- Lecturer Feedback
- User Profile Management

2.3 User Classes and Characteristics

User Class	Description	Technical Level
Administrator	Manages users and system activities	High
Student	Submits and improves innovative ideas	Medium
Lecturer	Reviews ideas and provides feedback	Medium

2.4 Operating Environment

Operating System

- Linux
- Windows

Browser Requirements

- Google Chrome
- Mozilla Firefox
- Microsoft Edge

Hardware Requirements

- Minimum 4GB RAM
- Internet-enabled Device

Network Requirements

- Stable Internet Connection

2.5 Design and Implementation Constraints

- Development period is limited to five weeks.
- React.js shall be used for frontend development.
- Spring Boot shall be used for backend development.
- PostgreSQL shall be used for database management.
- Open-source technologies shall be used throughout the project.

2.6 Assumptions and Dependencies

Assumptions

- Users have internet access.
- Users possess basic computer skills.
- Lecturers are willing to review student ideas.

Dependencies

- PostgreSQL Database
- Spring Boot Framework
- React.js Framework
- Internet Connectivity

3 Functional Requirements

3.1 Feature 1: User Authentication

FR-001: User Registration

- Description: The system shall allow new users to create accounts.
- Priority: High
- Input: Full Name, Email, Password
- Processing: Validate user information and store account details.
- Output: Successful registration message.
- Acceptance Criteria: User account is created successfully.

FR-002: User Login

- Description: The system shall allow registered users to log in.
- Priority: High
- Input: Email and Password
- Processing: Verify credentials.
- Output: Access to system dashboard.
- Acceptance Criteria: Valid users can log in successfully.

3.2 Feature 2: Idea Management

FR-003: Submit Idea

- Description: Students shall submit innovative ideas.
- Priority: High
- Input: Idea Title, Category, Description.
- Processing: Store idea in database.
- Output: Idea published successfully.
- Acceptance Criteria: Submitted idea appears in the system.

FR-004: View Ideas

- Description: Users shall browse available ideas.
- Priority: High
- Input: Search keyword or category.
- Processing: Retrieve ideas from database.
- Output: List of ideas.
- Acceptance Criteria: Users can view available ideas.

FR-005: Edit Idea

- Description: Students shall modify their ideas after receiving feedback.
- Priority: High
- Input: Updated idea information.
- Processing: Update stored idea.
- Output: Updated idea details.
- Acceptance Criteria: Changes are saved successfully.

3.3 Feature 3: Communication and Feedback

FR-006: Comment on Idea

- Description: Users shall comment on ideas.
- Priority: Medium
- Input: Comment text.
- Processing: Store comment.
- Output: Comment displayed under idea.
- Acceptance Criteria: Comment appears successfully.

FR-007: Lecturer Feedback

- Description: Lecturers shall provide feedback and recommendations.
- Priority: High
- Input: Feedback message.
- Processing: Save feedback.
- Output: Feedback displayed to students.
- Acceptance Criteria: Students can view lecturer feedback.

3.4 Feature 4: Idea Evaluation

FR-008: Vote for Idea

- Description: Users shall vote for innovative ideas.
- Priority: Medium
- Input: Vote action.
- Processing: Update vote count.
- Output: Updated vote total.
- Acceptance Criteria: Vote count increases correctly.

4 Non-Functional Requirements

4.1 Performance Requirements

- NFR-P01: The system shall respond to user requests within 3 seconds.
- NFR-P02: The system shall support at least 1000 concurrent users.
- NFR-P03: The system shall load pages within 5 seconds.
- NFR-P04: Search operations shall complete within 2 seconds.

4.2 Security Requirements

- NFR-S01: All passwords shall be encrypted using industry-standard encryption.
- NFR-S02: User sessions shall timeout after 30 minutes of inactivity.
- NFR-S03: Only authenticated users shall access protected resources.
- NFR-S04: User inputs shall be validated before processing.

4.3 Usability Requirements

- NFR-U01: The system shall be accessible on mobile devices.
- NFR-U02: New users shall complete basic tasks within 5 minutes without training.
- NFR-U03: Navigation shall be simple and user-friendly.
- NFR-U04: Error messages shall be clear and understandable.

4.4 Reliability Requirements

- NFR-R01**:** The system shall be available to users whenever internet connectivity is available
- NFR-R02: Data shall be backed up daily.
- NFR-R03: The system shall recover from failures without significant data loss.
- NFR-R04: Data consistency shall be maintained at all times.

5 User Stories

5.1 User Story Template

As a [type of user], I want [some goal] so that [some reason].

5.2 User Story 1

As a student, I have an innovative idea but I do not know how to make it a real solution. I want to submit my idea and receive guidance from lecturers so that I can improve and develop it into a practical project.

Acceptance Criteria

1. Given a logged-in student, when an idea is submitted, then the idea is saved successfully.
2. Given a submitted idea, when a lecturer reviews it, then feedback is provided.
3. Given lecturer feedback, when the student updates the idea, then changes are saved successfully.

5.3 User Story 2

As a lecturer, I want to review student ideas and provide recommendations so that students can improve their innovations and solve real-world challenges.

Acceptance Criteria

1. Given submitted ideas, when a lecturer selects an idea, then the idea details are displayed.
2. Given an idea review, when feedback is submitted, then the feedback is stored successfully.

6 Use Cases

6.1 Use Case 1: Submit Idea

Field	Description
Use Case ID	UC-001
Name	Submit Idea
Actor(s)	Student
Description	Student submits an innovative idea
Preconditions	Student is logged into the system

Main Flow

1. Student selects "Submit Idea".
2. System displays idea submission form.
3. Student enters idea details.
4. Student clicks Submit.
5. System validates information.
6. System stores the idea.
7. System displays confirmation message.

Alternative Flows

- Required fields are missing.
- System displays validation errors.

Postconditions

- Idea is stored successfully .

7 Requirements Traceability Matrix

Req ID	User Story	Test Case	Status
FR-001	US-001	TC-001	Pending
FR-002	US-002	TC-002	Pending
FR-003	US-003	TC-003	Pending
FR-004	US-004	TC-004	Pending
FR-005	US-005	TC-005	Pending
FR-006	US-006	TC-006	Pending
FR-007	US-007	TC-007	Pending
FR-008	US-008	TC-008	Pending

8 Approval

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